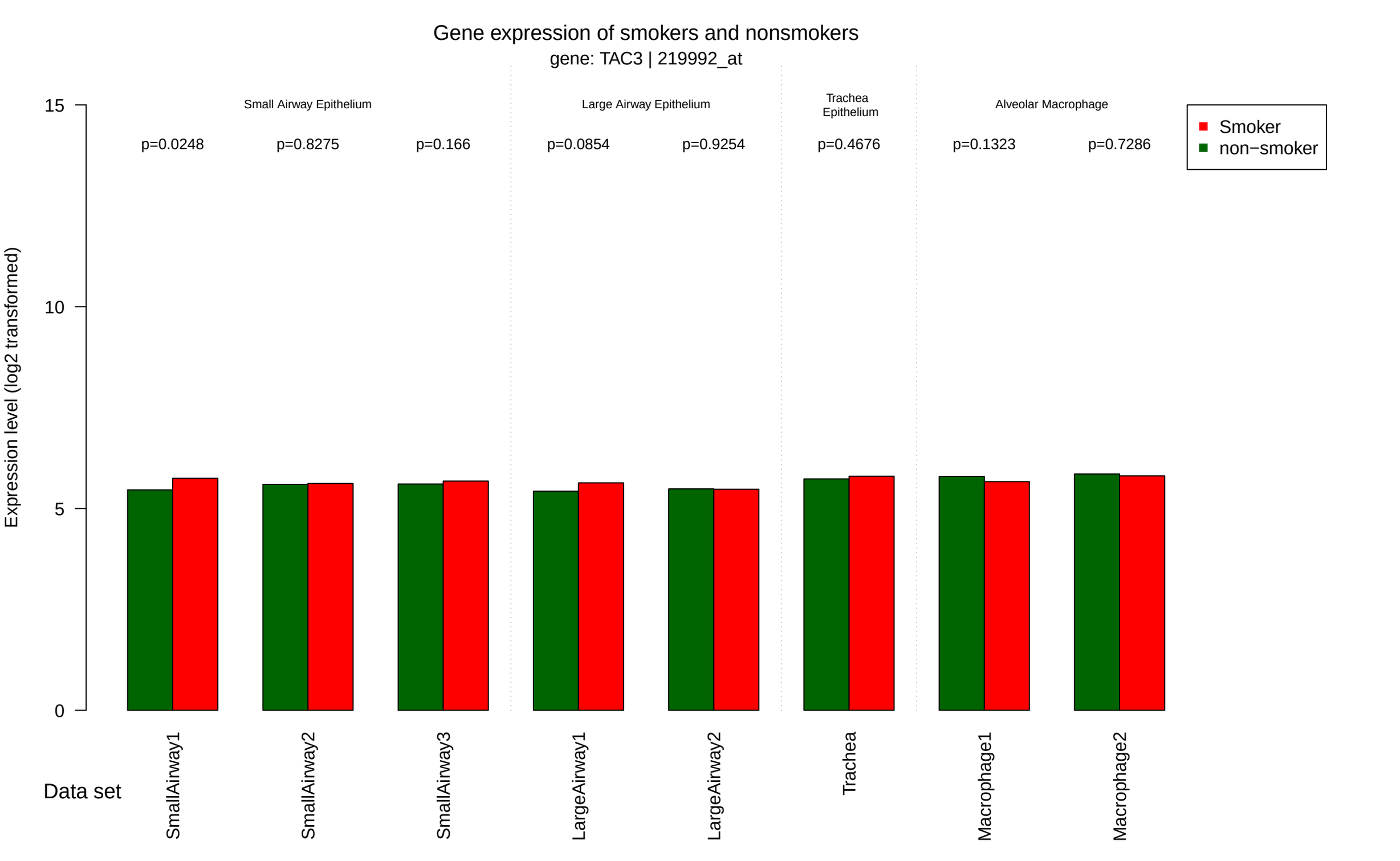
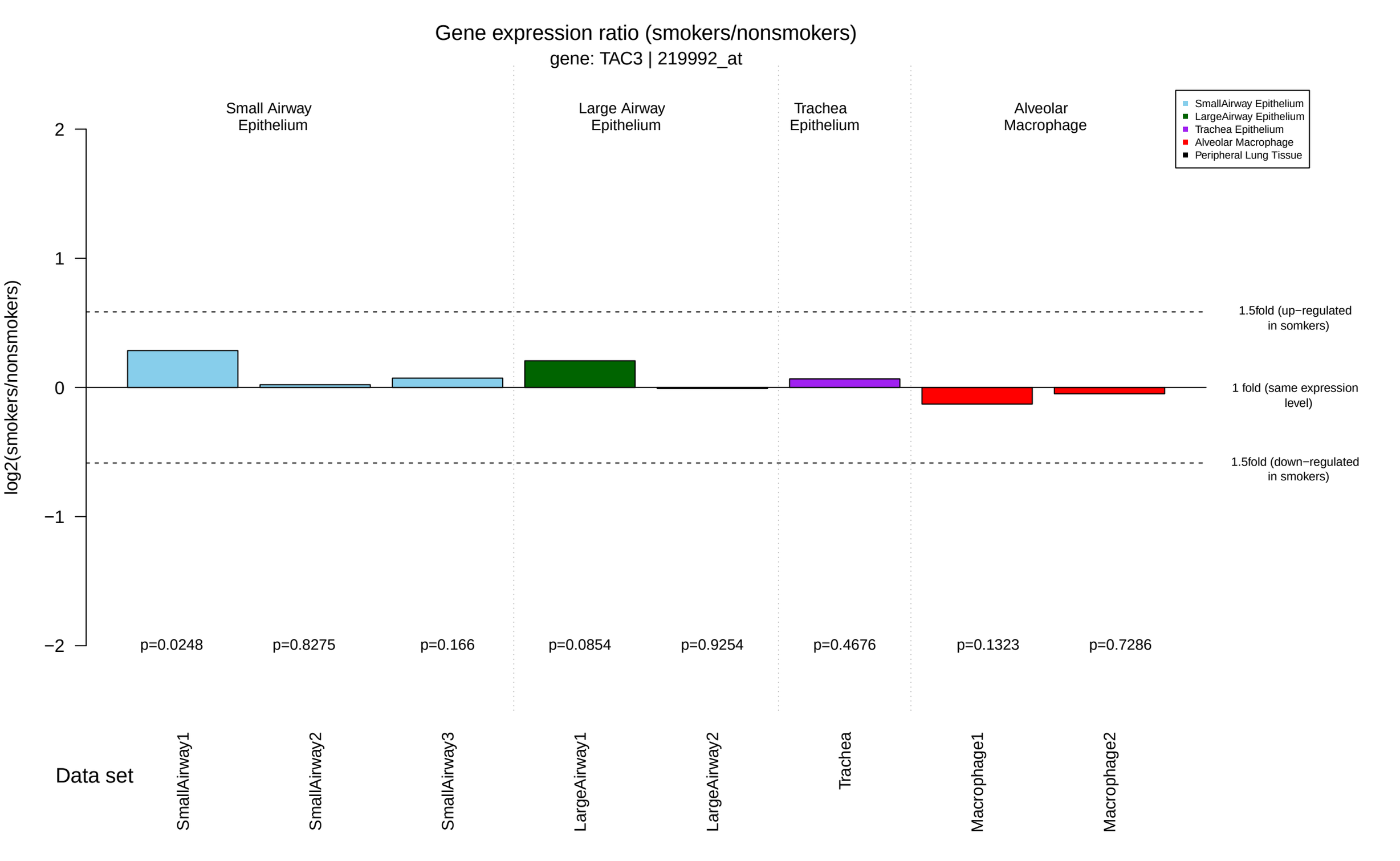
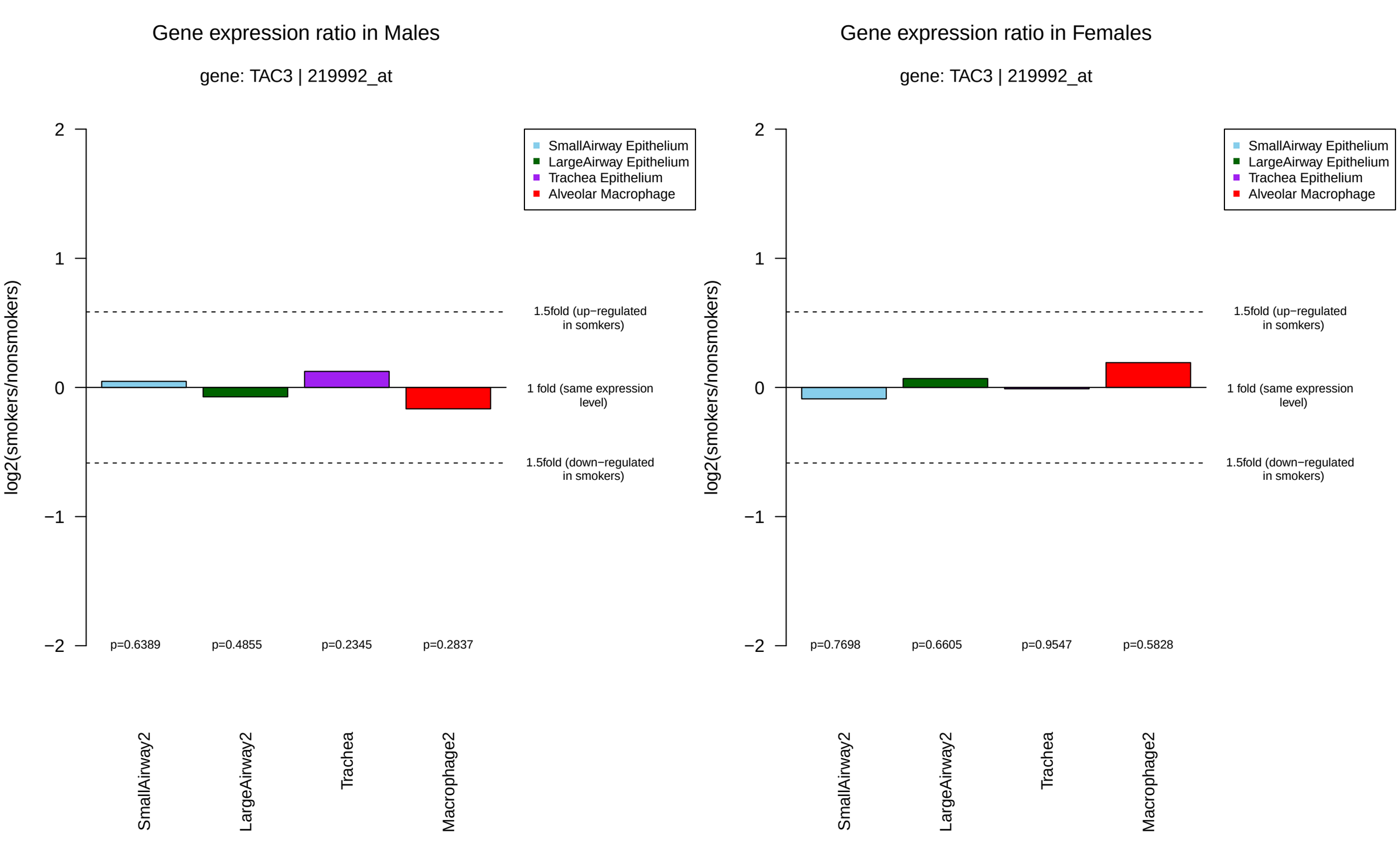
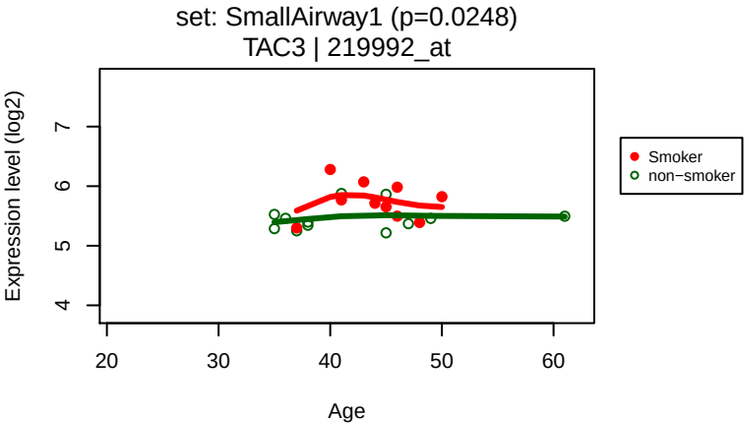
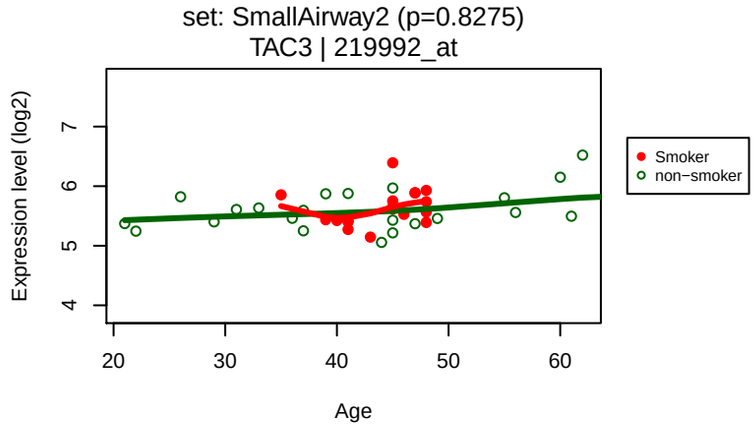
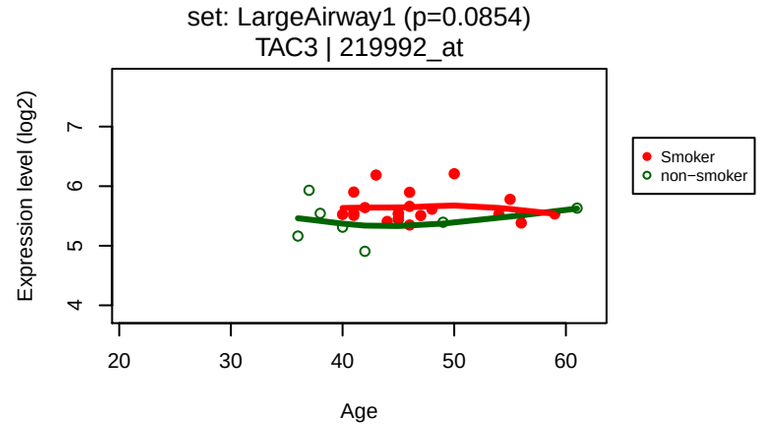

Gene expression of smokers and nonsmokers
gene: TAC3 | 219992_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0248
p=0.8275
p=0.166
p=0.0854
p=0.9254
p=0.4676
p=0.1323
p=0.7286
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

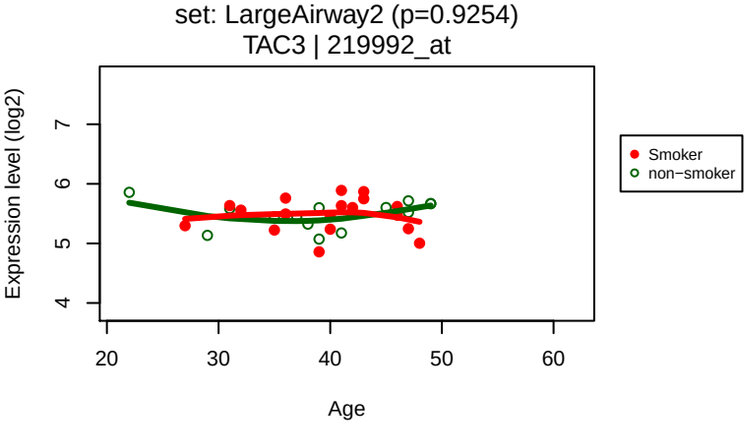

Gene expression ratio (smokers/nonsmokers)
gene: TAC3 | 219992_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0248
p=0.8275
p=0.166
p=0.0854
p=0.9254
p=0.4676
p=0.1323
p=0.7286
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

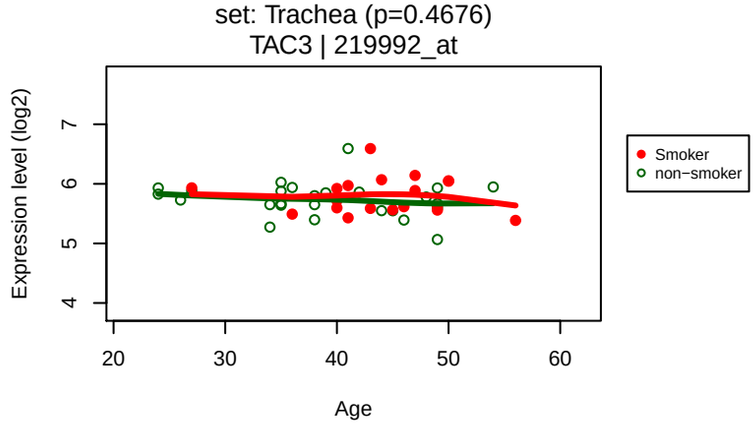

Gene expression ratio in Males
gene: TAC3 | 219992_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6389
p=0.4855
p=0.2345
p=0.2837
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAC3 | 219992_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7698
p=0.6605
p=0.9547
p=0.5828
SmallAirway2
LargeAirway2
Trachea
Macrophage2

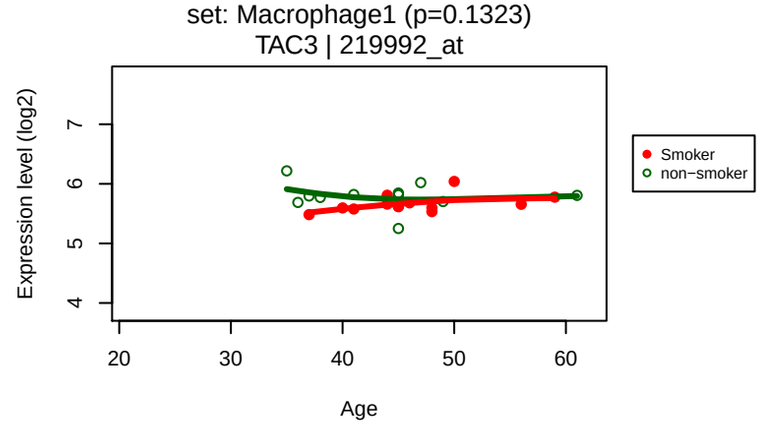

set: SmallAirway1 (p=0.0248)
TAC3 | 219992_at
Expression level (log2)
Age
Smoker
non−smoker

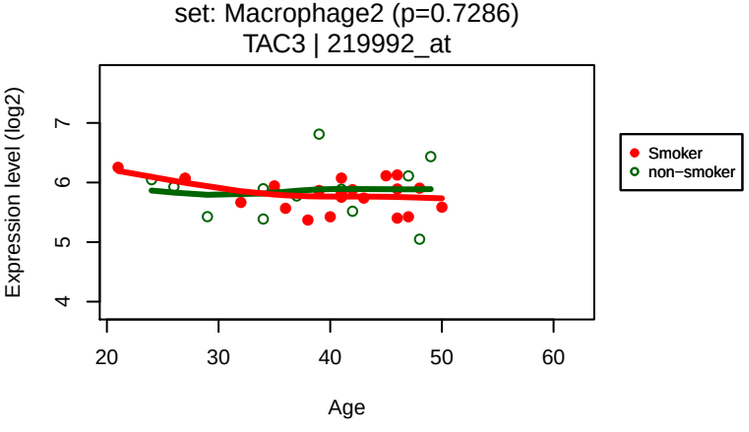

set: SmallAirway2 (p=0.8275)
TAC3 | 219992_at
Expression level (log2)
Age
Smoker
non−smoker

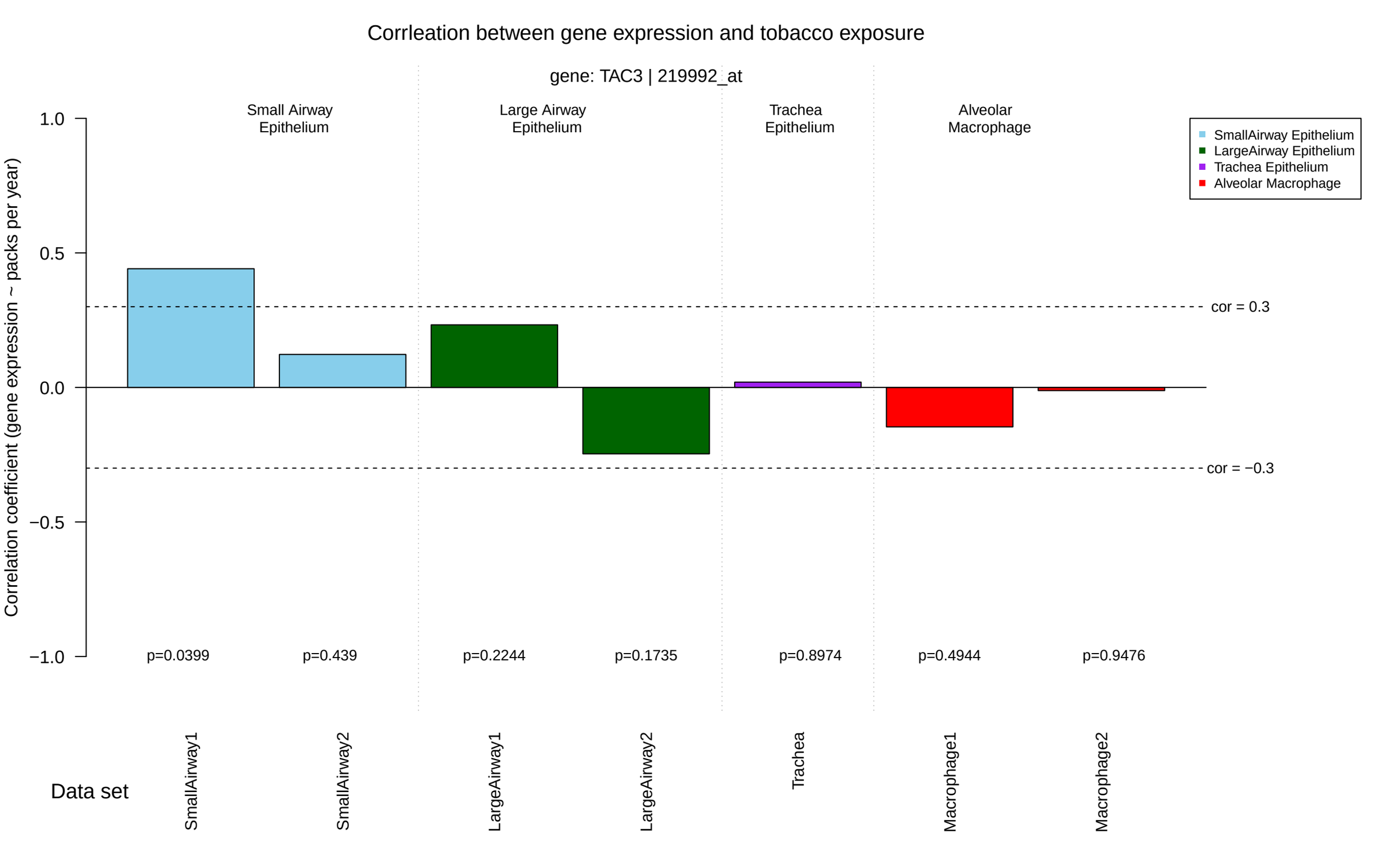

set: LargeAirway1 (p=0.0854)
TAC3 | 219992_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9254)
TAC3 | 219992_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4676)
TAC3 | 219992_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1323)
TAC3 | 219992_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7286)
TAC3 | 219992_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAC3 | 219992_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0399
p=0.439
p=0.2244
p=0.1735
p=0.8974
p=0.4944
p=0.9476
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2